FTP (FILE TRANSFER PROTOCOL)

File Transfer Protocol (FTP) is a standard network protocol used to transfer files between a client and a server over a TCP/IP network. The client connects to the FTP server, authenticates with a username and password, and can then upload, download, rename, or delete files, depending on the permissions granted. It uses a client-server model with two separate channels. The control connection (TCP port 21) carries commands and responses, and the data connection (TCP port 20 in active mode) carries the actual file data. This separation allows commands to be exchanged while files are being transferred.
It is widely used for website file management, software distribution, backups, and sharing large files within schools, offices, and organizations.

Advantages of FTS

- Allows large files and multiple files to be transferred quickly between computers over a network.
- Provides user authentication, so only authorized users can access the files stored on the server.
- Supports permission control, allowing different users to read, write, delete, or list files.
- Allows files to be centrally stored on a server, making backup and file management easier.
- Supports resuming interrupted transfers, which saves time when working with large files.
- Works across different operating systems and platforms, since it is a standard protocol.

Disadvantages of FTS

- Standard FTP sends usernames, passwords, and data in plain text, so it is not secure without additional protection such as FTPS or SFTP.
- Vulnerable to attacks such as packet sniffing and brute-force attacks when not properly secured.
- Requires a dedicated server and proper configuration of user accounts and permissions.
- Uses multiple connections, which can cause problems with firewalls and NAT.
- If the FTP server fails, no user can upload or download files until it is restored.
- File transfer speed depends on the server’s capacity and the network bandwidth.

Simulation / Design in Cisco Packet Tracer

This section presents the **FTP** network designed and simulated in Cisco Packet Tracer. The network consists of an **FTP** server and multiple PCs connected to a central switch. The server stores files and provides **FTP** service, while the PCs act as **FTP** clients that connect to the server to upload and download files. This arrangement demonstrates the working of FTP in a **centralized** network.

Network Diagram

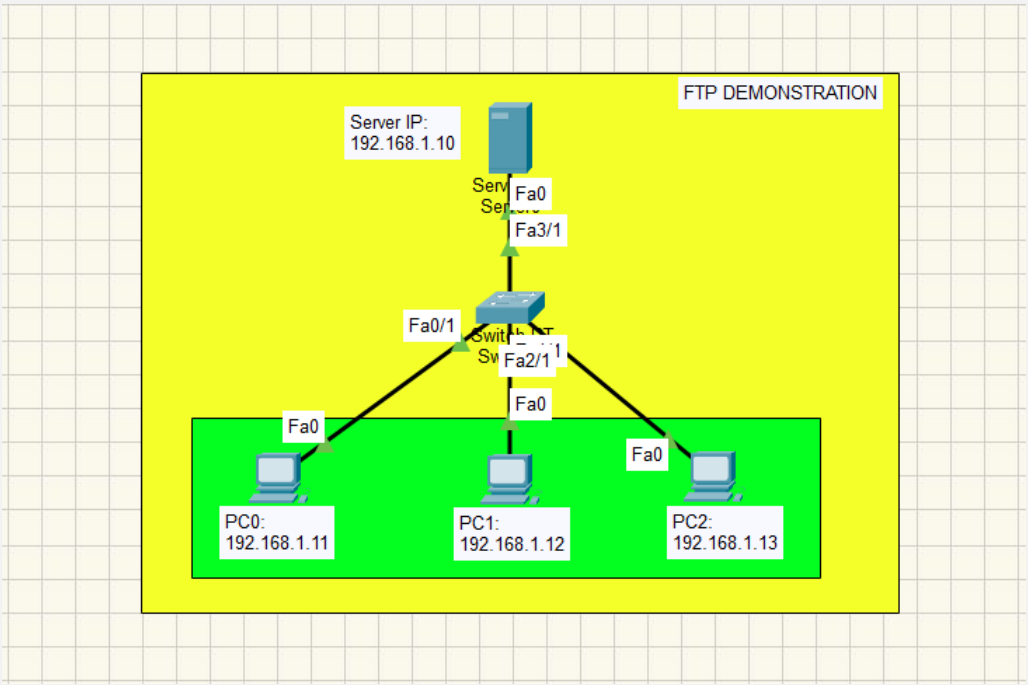


Figure 1: Star topology layout in Cisco Packet Tracer

Device Configuration



Figure 2.1: IP addressing / device configuration window



Figure 2.2: IP addressing / device configuration window

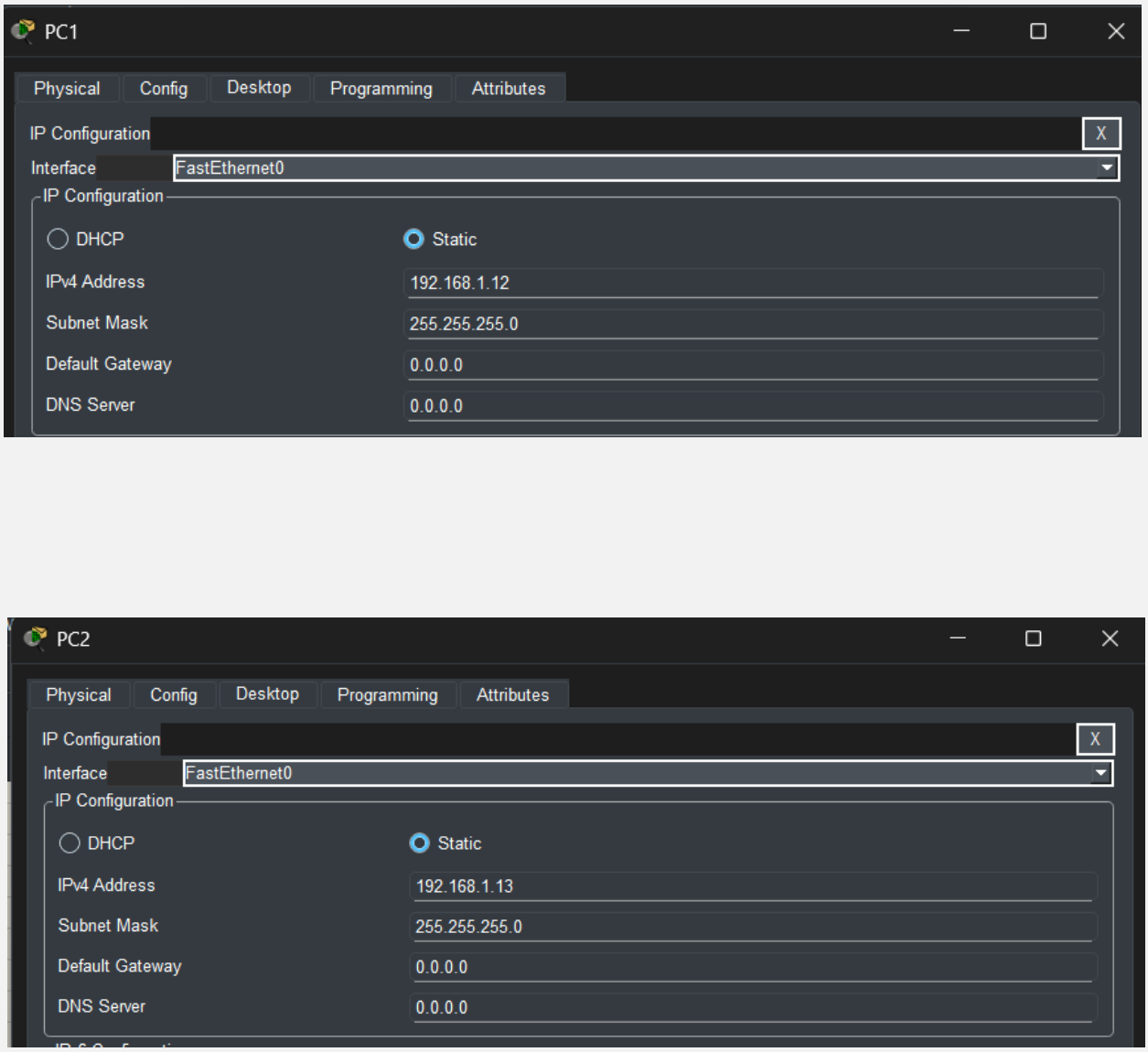


Figure 2.3: IP addressing / device configuration window

Connectivity Test (FTP/ Simulation Result)

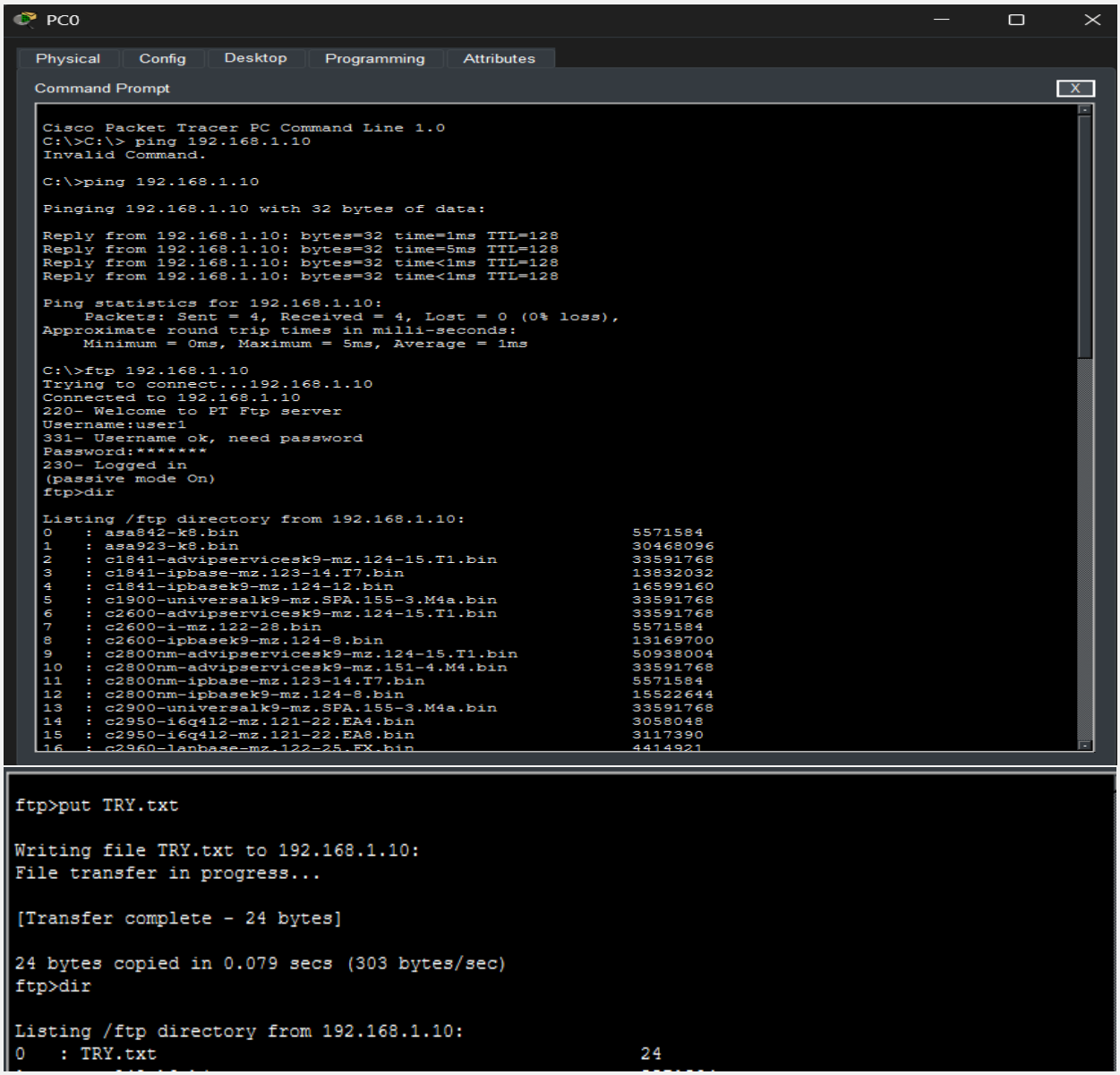


Figure 3.1: Successful FTP login and file transfer from the PC command prompt]

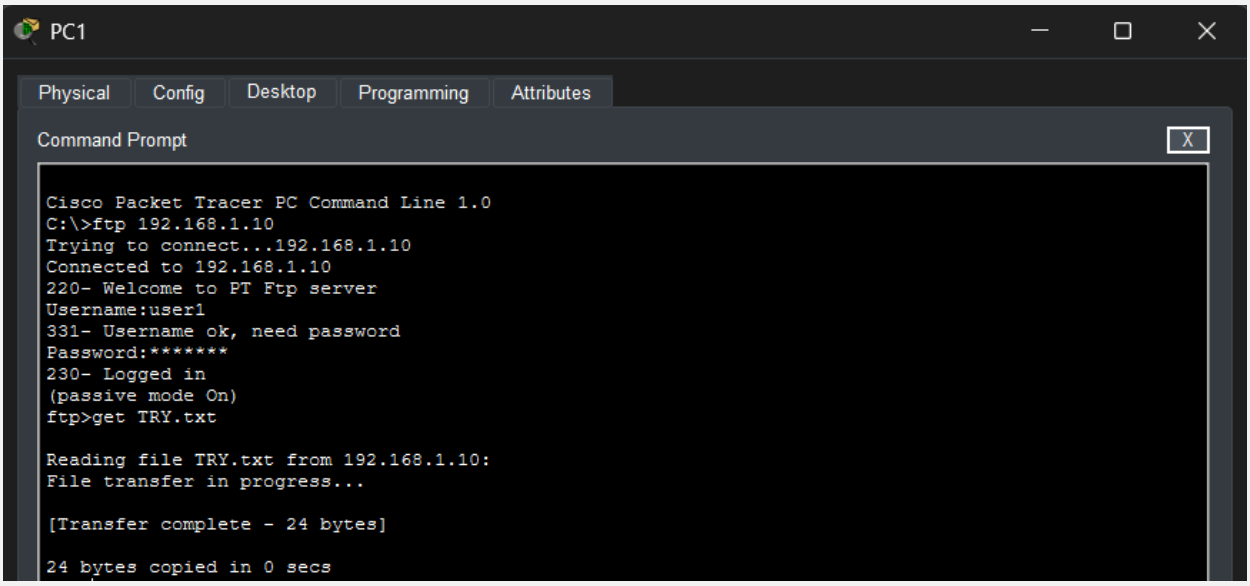


Figure 3.2: Successful packet simulation result between PC and FTP server

Process of Designing the Network in Packet Tracer

- Open Cisco Packet Tracer and create a new blank project to begin designing the FTP network.
- Build a minimal topology by placing one 2960 switch at the center of the workspace. From the End Devices category, place one Server-PT, which will act as the FTP server, and two PC-PTs, named PC0 and PC1, which will act as FTP clients.
- Select Copper Straight-Through cable from the Connections category and connect the Server, PC0, and PC1 individually to the switch. Wait until the link lights turn green, which shows that the physical connections are active.
- Configure the network settings of each device by opening Desktop → IP Configuration. All devices are placed in the same network, so no default gateway is required.

Device	IP Address	Subnet Mask
Server	192.168.1.10	255.255.255.0
PC0	192.168.1.11	255.255.255.0
PC1	192.168.1.12	255.255.255.0

- Configure the FTP service on the server by opening Server → Services → FTP and setting the service to On.
- A default user account with the username cisco and password cisco already exists, with full permissions (Write, Read, Delete, Rename, and List). This account can be used directly, or a new account can be created.
- To create a new account, enter a username (for example, user1) and a password (for example, pass123), tick the permissions required, and click the + button to add the user. For a read-only user, tick only the Read and List permissions.
- Create a file on PC0 to upload. Open Desktop → Text Editor on PC0, type some text, and save the file using File → Save as with the name hello.txt.
- Test connectivity before using FTP. Open the Command Prompt on PC0 and enter the command ping 192.168.1.10. Four successful replies confirm that PC0 can reach the FTP server.
- Demonstrate the file upload from PC0 to the server. In the Command Prompt of PC0, connect to the server using the command ftp 192.168.1.10. Enter the username cisco and the password cisco when prompted.
- After logging in, type dir to list the files currently stored on the server. Then type put hello.txt to upload the file to the server. A message such as “Transfer complete” confirms the upload.
- Type dir again. The list now includes hello.txt, which proves that the file has been successfully stored on the server.
- Demonstrate the file download from the server to PC1. Open the Command Prompt on PC1 and connect using the command ftp 192.168.1.10. Enter the username cisco and the password cisco.
- Type dir to confirm that hello.txt is available on the server. Then type get hello.txt to download the file to PC1. A transfer complete message confirms the download.
- Type quit to close the FTP session and return to the normal command prompt.
- Screenshots of the topology, IP configurations, FTP service settings, ping result, and the upload and download commands are captured as evidence of the completed setup.
- Finally, the completed Packet Tracer project is saved as a .pkt file, along with the required documentation and screenshots for submission.

Conclusion

The **FTP** network was successfully designed and configured in Cisco Packet Tracer, demonstrating how files can be shared between a server and multiple clients. The experiment involved configuring an FTP server, creating a user account, assigning IP addresses, and transferring files from PCs. Connectivity and file transfer were verified using the ping command, FTP commands, and Simulation Mode. This experiment reinforced the practical understanding of FTP, client-server communication, and file sharing over a network.